When Models Agree: Inter-Model Consensus Predicts Correctness for Facts and Math, but Much Less for Ethics

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Abstract

LLM juries, ensemble labeling, and synthetic-data filters all rely on the same assumption: when several independent language models agree on an answer, the answer is probably correct. We test how far this assumption holds and whether it is weaker for ethical judgement, where models may share biases. We run 8 open-weight instruction models from 7 developer families on 3,500 items from 7 tasks: factual QA, math, two ETHICS subsets, and a binary factual control matched to the ethics answer format. The agreement rate (the fraction of models giving the plurality answer) is a strong label-free correctness signal on factual and math tasks (AUROC 0.78–0.91, point-biserial r 0.37–0.61). It is clearly weaker on ethics (AUROC 0.63–0.71, r 0.20–0.33; pooled factual minus ethics ΔAUROC $= +0.12$, 95% CI $[0.08, 0.17]$). However, the binary factual control reaches only AUROC 0.735, which is indistinguishable from ETHICS commonsense, so answer format explains about half of the gap. The plurality vote beats the best single model only on math ($+2.0$ to $+2.4$ pp). It is 1.8–2.4 pp below the best model on factual QA, a difference that is not significant, and significantly below it on ethics (-4.8 pp, Holm $p = 0.008$). On ethics, model errors are more correlated than on the control (error ϕ 0.45–0.49 vs. 0.30), and the vote's deficit grows as models are added. Cross-model agreement is therefore a useful triage signal for factual and math answers, but it should not replace human labels for moral judgements: 10–17% of unanimous ETHICS verdicts contradict the human label.

1 Introduction

When we cannot check an answer, we often ask a second opinion. The same intuition now runs through many LLM pipelines. Panels of models act as evaluators [Verga et al., 2024], ensembles vote on labels and answers [Li et al., 2024, Jiang et al., 2023], and agreement across samples is used to detect hallucinations and to pick answers [Wang et al., 2023]. All of these practices assume that *consensus implies correctness*.

This assumption matters most where labels are scarce and decisions are value-laden. If model errors were independent, agreement among several models would be strong evidence of a correct answer. But models are trained on overlapping data with similar objectives, and recent work shows that their errors are strongly correlated [Goel et al., 2025, Kim et al., 2025]. Shared errors are especially worrying for ethical judgement. There, models may share leanings learned from pre-training data and alignment, and the reference labels themselves reflect one group's norms [Hendrycks et al., 2021a]. In that setting, a confident consensus could be confidently wrong.

Prior work has studied pieces of this question in isolation. On one hand, self-consistency shows that agreement *within* one model predicts correctness on math [Wang et al., 2023], and multi-agent voting reports accuracy gains [Li et al., 2024]. On the other hand, studies of correlated errors measure similarity between models but do not ask how well agreement works as a correctness signal in each

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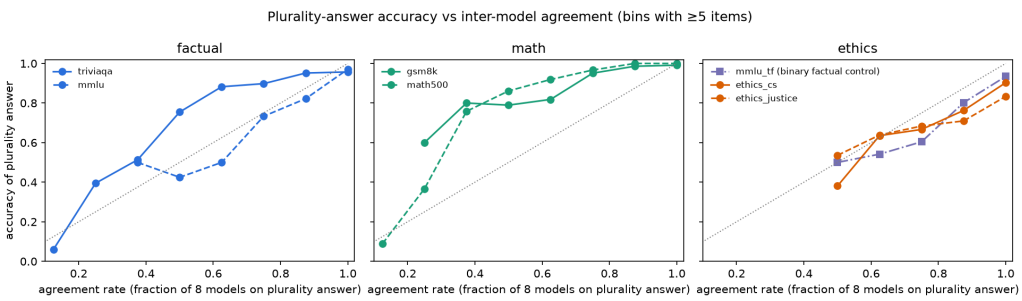


Figure 1: Accuracy of the plurality answer as a function of inter-model agreement (fraction of 8 models giving the plurality answer; bins with $\geq$5 items; dotted line is the diagonal). On factual and math tasks, accuracy rises steeply with agreement and approaches 1 at full agreement. On ethics, items concentrate at high agreement, yet accuracy stays at 0.83–0.90 even when all 8 models agree. The binary factual control (MMLU-TF, purple) follows a similar curve to the ethics tasks, which shows that part of the weakness comes from the binary answer format.

domain. What is missing is one controlled comparison on *the same model pool* that measures (a) how strongly agreement tracks correctness, (b) whether a majority vote beats the best single model, and (c) how well agreement separates right from wrong answers without labels, across factual, math and ethics tasks.

Our main question is: **how much does inter-model agreement tell us about correctness, and is it a weaker signal for ethics?** We hypothesize that agreement correlates with correctness at $r \in [0.4, 0.6]$ on factual and math tasks but at $r < 0.2$ on ethics, that a vote of five or more models beats the best model by 3–7 pp on factual QA but not on ethics, and that agreement alone reaches AUROC 0.70–0.80 on factual tasks. To test this, we run 8 open-weight instruction models from 7 developer families on 7 tasks of 500 items each. The tasks cover open-ended trivia (TRIVIAQA), multiple-choice knowledge (MMLU), math word problems (GSM8K), competition math (MATH-500) and two ETHICS subsets. We also add a control: MMLU-TF, a *binary* true/false version of MMLU. It matches the two-way answer format of ETHICS, which lets us separate the effect of the ethics domain from the effect of a binary answer space.

We find that agreement is a strong label-free signal on factual and math tasks (AUROC 0.78–0.91), and that the plurality answer is correct 96–100% of the time when all 8 models agree on open-ended tasks (figure 1). On ethics the signal is clearly weaker (AUROC 0.63–0.71). However, about half of this gap comes from answer format: the binary factual control (AUROC 0.735) is statistically indistinguishable from ETHICS commonsense (0.711, $p = 0.48$). The domain-specific part of the story lies in the vote. On ethics, model errors are more correlated than on the control (mean pairwise ϕ 0.45–0.49 vs. 0.30). The plurality vote is significantly worse than the best model (-4.8 pp, Holm $p = 0.008$), and this deficit grows from -1.3 pp to -4.8 pp as the ensemble grows from 3 to 8 models. Contrary to our hypothesis, voting also fails to beat the best model on factual QA (-1.8 to -2.4 pp, not significant) and helps only on math ($+2.0$ to $+2.4$ pp).

In summary, our contributions are:

- We conduct a controlled, cross-domain study of inter-model agreement as a correctness signal on 3,500 items, with 8 models from 7 families answering every item.
- We introduce a format-matched binary factual control and show that roughly half of the factual-vs-ethics gap in AUROC is due to answer format rather than domain.
- We show that agreement-based ranking and majority voting behave differently. Agreement ranks items well everywhere, but the vote beats the best model only on math, and on ethics its deficit grows with ensemble size, as correlated errors would predict.
- We analyze unanimous-but-wrong cases and find that ethics consensus errors combine crowd-label noise with systematic shared model leanings, while factual consensus errors are rare: only 0.4–2.2% of open-answer and math items have $\geq$75% agreement on a wrong answer.

Section 2 reviews related work, section 3 describes the setup, section 4 presents results, section 5 discusses implications and limitations, and section 6 concludes.

2 Related Work

Agreement within a single model. Self-consistency samples several reasoning chains from one model and takes the most common final answer [Wang et al., 2023]. It improves math accuracy, and the vote margin predicts correctness. Related methods use disagreement between samples to flag hallucinations [Manakul et al., 2023, Kuhn et al., 2023], or use the model's own token probabilities as a confidence score [Kadavath et al., 2022]. Unlike these works, we measure agreement *across* models with different training pipelines, using a single greedy answer per model. We also compare it directly with each model's own sequence probability as a single-model baseline.

Ensembles and LLM panels. Classical ensemble theory says that voting helps when members are accurate and their errors are diverse [Dietterich, 2000, Kuncheva and Whitaker, 2003]. For LLMs, Li et al. [2024] report that accuracy grows with the number of sampled agents, and LLM-Blender ranks and fuses outputs from several models [Jiang et al., 2023]. Verga et al. [2024] replace a single large judge with a panel of smaller models from different families and report better agreement with humans. Building on this line of work, we test whether a plurality vote over a diverse pool beats the *best* member, not just the average one, and how the answer changes across domains and ensemble sizes.

Correlated errors across models. Recent studies find that LLM errors are strongly correlated, and more so for more capable models or models that share a developer or base model [Goel et al., 2025, Kim et al., 2025]. Goel et al. [2025] argue that this similarity weakens AI oversight, because a judge model tends to favor models that resemble it. We study the consequence of correlated errors for consensus as a correctness signal, and we compare error correlation across factual, math and ethics tasks under a format-matched control.

Machine ethics benchmarks. ETHICS [Hendrycks et al., 2021a] collects crowd-labeled moral judgements about everyday scenarios. Its "hard" test split is adversarially filtered, and some of its items are ambiguous. Other datasets record several human judgements per item, e.g., Scruples [Lourie et al., 2021] and MoralChoice [Scherrer et al., 2023], which allows a comparison with human disagreement. We use ETHICS because it has a single reference label per item, which lets us define correctness in the same way as for factual tasks. We then examine where model consensus and the crowd label part ways.

Selective prediction. Abstaining on low-confidence inputs trades coverage for accuracy [Geifman and El-Yaniv, 2017]. We evaluate inter-model agreement as a label-free abstention rule and report coverage–accuracy curves per task.

3 Methodology

3.1 Problem formulation

Let $\mathcal{M} = \{m_1, \ldots, m_M\}$ be a pool of $M = 8$ models, and let x be a test item with reference answer $y^\star$. Each model gives one answer $\hat{y}_j(x)$. We group the answers into equivalence classes using a task-specific equivalence relation $\equiv$ (section 3.3). An answer that cannot be parsed counts as wrong and casts no vote. Let $G_1, \ldots, G_K$ be the resulting groups. The **plurality answer** $\hat{y}(x)$ is the answer of the largest group, with ties broken by a fixed model order. The **agreement rate** is

$$a(x) = \frac{1}{M} \max_k |G_k| \in \left\{ \tfrac{1}{8}, \tfrac{2}{8}, \ldots, 1 \right\}. \tag{1}$$

We call an item *unanimous* if $a(x) = 1$. Let $c(x) = \mathbf{1}[\hat{y}(x) \equiv y^\star]$ be the correctness of the plurality answer. Our question is how well $a(x)$ predicts $c(x)$ without access to $y^\star$.

3.2 Models

API keys for commercial models were not available, so we run open-weight instruction models locally with vLLM [Kwon et al., 2023] on shared NVIDIA RTX A6000 GPUs. Each GPU had only 6–10 GB of free memory, so every model uses a 4-bit weight-only checkpoint (AWQ [Lin et al., 2024] or GPTQ [Frantar et al., 2023], W4A16). All models therefore share the same precision. The pool contains 8 models from 7 developer families (table 1). We decode greedily (temperature 0, seed 42) with each model's own chat template, a single user turn, and no system prompt. Qwen3 runs in

Key	4-bit checkpoint	Family
qwen2.5-7b	Qwen2.5-7B-Instruct-AWQ	Alibaba Qwen
qwen3-8b	Qwen3-8B-AWQ (non-thinking)	Alibaba Qwen
llama3.1-8b	Meta-Llama-3.1-8B-Instruct-AWQ-INT4	Meta
gemma2-9b	gemma-2-9b-it-AWQ-INT4	Google
mistral-7b	Mistral-7B-Instruct-v0.3-AWQ	Mistral
falcon3-7b	Falcon3-7B-Instruct-AWQ	TII
olmo3-7b	Olmo-3-7B-Instruct-gptq-w4a16-g128	AI2
phi3-mini	Phi-3-mini-4k-instruct-gptq-4bit	Microsoft

Table 1: The model pool: 8 instruction-tuned models (3.8B–9B parameters) from 7 developer families, all run at the same 4-bit weight-only precision.

Dataset	Domain	Split	Answer format	Equivalence
TRIVIAQA	factual	rc.nocontext, val.	open short answer	normalized string
MMLU	factual	test, moral_* excluded	4-way MC	letter
GSM8K	math	test	CoT $\rightarrow$ number	numeric
MATH-500	math	test	CoT $\rightarrow$ `\boxed{}`	normalized LaTeX
ETH-CS	ethics	commons., test_hard, short	wrong / acceptable	label word
ETH-JUSTICE	ethics	justice, test_hard	(un)reasonable	label word
MMLU-TF (control)	factual	MMLU test, disjoint items	true / false	label word

Table 2: Tasks: TRIVIAQA [Joshi et al., 2017], MMLU [Hendrycks et al., 2021b], GSM8K [Cobbe et al., 2021], MATH-500 [Hendrycks et al., 2021c, Lightman et al., 2024] and ETHICS [Hendrycks et al., 2021a]. CoT = chain of thought; MC = multiple choice. Each has 500 items (491 for TRIVIAQA after removing duplicates). MMLU-TF is a binary factual control whose answer format matches ETHICS.

non-thinking mode. The maximum number of new tokens is 8 for MMLU, ETHICS and MMLU-TF, 32 for TRIVIAQA, 768 for GSM8K, and 1,536 for MATH-500. The full set of generations took about 1.4 GPU-hours.

3.3 Tasks

We sample 500 items per task with seed 42 (table 2). TRIVIAQA's validation split repeats 9 question ids, so we remove the duplicates and keep $n = 491$. Factual and ethics tasks ask for a direct answer. Math tasks ask for a chain of thought followed by a final answer.

Answer equivalence. For MMLU, the answer is a letter. For GSM8K, we compare numbers. For MATH-500, we compare LaTeX strings after normalizing fractions, `\left`/`\right`, `\text{}` and similar markup. For TRIVIAQA, we apply SQuAD-style normalization and strip accents and plurals. A TRIVIAQA answer is correct if it contains a gold alias. Two TRIVIAQA answers agree if they match exactly, or if one contains the other and the shorter one has at least two tokens. The two-token rule keeps agreement close to transitive: with one-token containment, an answer like "eggs" would match every answer ending in "eggs".

Binary factual control. ETHICS answers are binary, while TRIVIAQA and MATH-500 have open answer spaces. A binary answer space produces high agreement by chance and allows only coarse agreement levels. To separate this format effect from the domain effect, we build MMLU-TF. Each item shows an MMLU question with one proposed answer and asks whether the answer is true or false. The proposed answer is the gold option for 50% of items and a random distractor for the rest. The MMLU-TF questions do not overlap with the main MMLU sample.

Most model–task pairs have parse-failure rates of 0–2.6%. MATH-500 is the exception, with 3–22%: Gemma-2 often omits `\boxed{}`, and Qwen3 and Llama sometimes reach the token limit.

	Factual		Math		Ethics		Control
Model	TRIVIAQA	MMLU	GSM8K	MATH-500	ETH-CS	ETH-JUSTICE	MMLU-TF
qwen2.5-7b	0.530	0.682	0.916	0.720	0.776	0.700	0.700
qwen3-8b	0.521	**0.728**	0.928	0.756	0.720	0.742	0.700
llama3.1-8b	0.654	0.580	0.844	0.406	0.720	0.690	0.678
gemma2-9b	**0.725**	0.692	0.882	0.428	**0.830**	**0.788**	**0.724**
mistral-7b	0.682	0.530	0.548	0.118	0.730	0.684	0.670
falcon3-7b	0.509	0.674	0.872	0.686	0.774	0.708	0.686
olmo3-7b	0.401	0.580	0.922	0.748	0.658	0.646	0.652
phi3-mini	0.491	0.628	0.820	0.378	0.774	0.692	0.688
Mean model	0.564	0.637	0.842	0.530	0.748	0.706	0.687
Plurality vote (8)	0.701	0.710	**0.952**	**0.776**	0.782	0.746	0.718

Table 3: Accuracy of each model and of the 8-model plurality vote. Best result per column in **bold**. The vote is the best predictor only on the two math tasks. On factual and ethics tasks a single model (Gemma-2 or Qwen3) beats it, although the vote is always well above the mean model.

3.4 Metrics and baselines

Agreement as a correctness signal. We report the point-biserial correlation r between $a(x)$ and $c(x)$, and the AUROC of $a(x)$ as a score for predicting $c(x)$. We also report a **leave-one-out (LOO)** variant that scores each model's own answer. For model m_j, the score is the fraction of the other 7 models that agree with $\hat{y}_j(x)$. We compute its AUROC for predicting whether $\hat{y}_j(x)$ is correct and average over the 8 models. As a single-model, label-free baseline, we use each model's own **sequence probability** for its answer. We report this baseline only on the direct-answer tasks, because the probability of a long chain of thought is not a meaningful confidence score.

Vote versus the best model. Our main comparison is plurality-vote accuracy minus the accuracy of the best single model. We pick the best model on the test set itself, which is an oracle choice that favors the baseline. We also report three variants: (a) the best model chosen on a random half of the items and scored on the other half, averaged over 200 splits; (b) the mean, over all 56 five-model subsets, of the vote minus the best model in that subset; (c) a sweep over ensemble size $k = 3, \ldots, 8$, averaging over all subsets of each size.

Error structure. We compute the mean pairwise ϕ correlation between the error indicators of different models. We also compute $P(\text{same wrong answer} \mid \text{both wrong})$, and the share of items with $a(x) \geq 0.75$ (at least 6 of 8 models) whose plurality answer is wrong.

Statistics. We compute 95% percentile bootstrap confidence intervals with 1,000 resamples, or 2,000 for group differences. We compare the vote with the best model using an exact McNemar test, with Holm correction across the 7 datasets. Domain differences in r and AUROC use a two-sample bootstrap; pooled domain estimates concatenate the items of the tasks in that domain.

4 Results

We first report per-model accuracy (section 4.1). We then ask whether agreement predicts correctness (section 4.2), whether the ethics gap is caused by domain or by format (section 4.3), whether voting beats the best model (section 4.4), and why consensus fails when it does (section 4.5).

4.1 Model accuracy

Table 3 shows that the pool is uneven, and that a different model leads on each kind of task. Gemma-2 is best on TRIVIAQA (0.725) and on both ethics tasks. Qwen3 is best on MMLU (0.728), GSM8K (0.928) and MATH-500 (0.756). The gap between the best model and the mean model is large on factual tasks: 16.1 pp on TRIVIAQA and 9.1 pp on MMLU. On MATH-500, accuracy ranges from 0.118 (Mistral) to 0.756 (Qwen3). This unevenness shapes the voting results in section 4.4.

Dataset	Domain	Agreement as a correctness signal				Consensus statistics			
		r [95% CI]	AUROC [95% CI]	LOO AUROC	Self-prob. AUROC	Error ϕ	Unanimous (%)	Acc. if unanimous	$\geq$75% agree, wrong (%)
TRIVIAQA	factual	0.548 [0.48, 0.61]	0.845 [0.80, 0.88]	0.888	0.803	0.50	14.3	0.957	2.2
MMLU	factual	0.446 [0.37, 0.52]	0.781 [0.74, 0.82]	0.845	0.717	0.40	27.4	0.971	8.0
GSM8K	math	0.368 [0.24, 0.48]	0.849 [0.75, 0.93]	0.951	–	0.36	45.0	0.991	1.4
MATH-500	math	**0.612** [0.56, 0.66]	**0.910** [0.88, 0.94]	**0.969**	–	0.40	6.6	**1.000**	**0.4**
ETH-CS	ethics	0.327 [0.24, 0.41]	0.711 [0.66, 0.76]	0.756	0.602	0.45	49.4	0.903	13.8
ETH-JUSTICE	ethics	0.201 [0.11, 0.29]	0.626 [0.58, 0.68]	0.691	0.596	0.49	46.8	0.833	19.4
MMLU-TF	control	0.373 [0.30, 0.45]	0.735 [0.69, 0.78]	0.765	0.587	0.30	30.4	0.934	13.4

Table 4: Agreement among 8 models as a label-free correctness signal. r and AUROC measure how well the agreement rate predicts the correctness of the plurality answer. LOO AUROC scores each model's answer by the fraction of the other 7 models that agree with it; self-prob. AUROC uses the model's own sequence probability (both averaged over the 8 models; self-probability is reported for direct-answer tasks only). The last column is the share of all items on which at least 6 of 8 models agree on a wrong answer. Best signal in **bold**. Agreement is strongest on math and weakest on ethics; LOO agreement beats self-probability on every task where both are available.

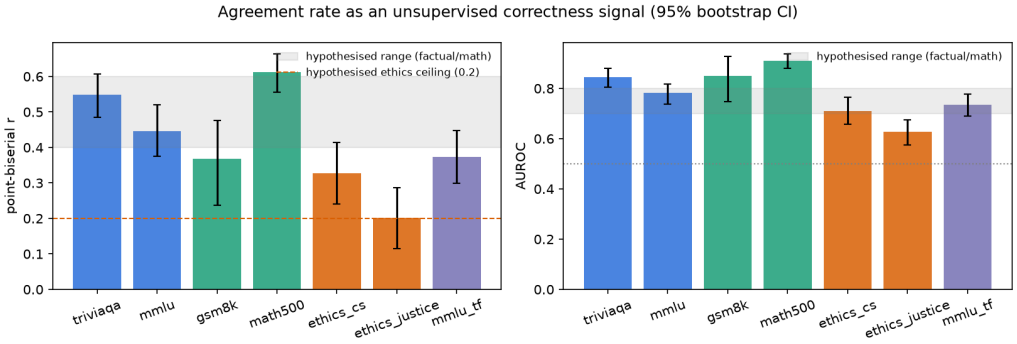


Figure 2: Point-biserial r (left) and AUROC (right) of the agreement rate for predicting plurality-answer correctness, with 95% bootstrap CIs. Gray bands mark the hypothesized ranges for factual and math tasks; the dashed line marks the hypothesized ethics ceiling $r = 0.2$. Factual and math tasks reach AUROC 0.78–0.91, while ethics tasks reach 0.63–0.71. The binary factual control (MMLU-TF, purple) lies close to the ethics tasks.

4.2 Agreement is a strong signal for facts and math, and a weaker one for ethics

Factual and math. Table 4 and figure 2 show that agreement predicts correctness well on all four factual and math tasks. AUROC is 0.845 on TRIVIAQA, 0.781 on MMLU, 0.849 on GSM8K and 0.910 on MATH-500. Pooled by domain, factual tasks reach $r = 0.48$ $[0.43, 0.53]$ and AUROC 0.79 $[0.76, 0.82]$, and math tasks reach $r = 0.58$ $[0.53, 0.63]$ and AUROC 0.91 $[0.89, 0.94]$. Full agreement is highly reliable. When all 8 models agree, the plurality answer is correct 95.7% of the time on TRIVIAQA, 97.1% on MMLU, 99.1% on GSM8K and 100% on MATH-500. GSM8K has a lower r (0.368) than the other tasks, even though its AUROC is high. The reason is that the vote is right on 95% of GSM8K items, and so few errors leave little variance for r to capture.

Ethics. On ethics, the signal is clearly weaker. ETH-CS reaches AUROC 0.711 and ETH-JUSTICE 0.626, and the pooled ethics domain reaches $r = 0.26$ $[0.20, 0.32]$ and AUROC 0.67 $[0.63, 0.70]$. Both the factual and math domains are significantly above ethics (table 5). The gap is $+0.124$ AUROC for factual tasks and $+0.246$ for math tasks, both with $p < 0.001$. Agreement on ethics is *high* but not very *informative*. Mean agreement is 0.87 on ethics, against 0.58–0.75 on TRIVIAQA and MMLU, and about half of ethics items are unanimous. Yet unanimous ethics verdicts are wrong 9.7% (ETH-CS) and 16.7% (ETH-JUSTICE) of the time (figure 1).

Other models are a better confidence signal than a model's own probability. The LOO variant scores each model's answer by how many of the other 7 models agree with it. It beats the model's own sequence probability on every direct-answer task: 0.888 vs. 0.803 on TRIVIAQA, 0.845 vs. 0.717 on MMLU, 0.756 vs. 0.602 on ETH-CS, 0.691 vs. 0.596 on ETH-JUSTICE, and 0.765 vs.

Comparison	ΔAUROC [95% CI]	Δr [95% CI]
Factual (open + MC) − ethics	+0.124 [+0.078, +0.172]***	+0.211 [+0.134, +0.292]***
Math − ethics	+0.246 [+0.201, +0.291]***	+0.319 [+0.239, +0.395]***
Factual (open + MC) − MMLU-TF	+0.056 [+0.007, +0.108]*	+0.102 [+0.012, +0.195]*
MMLU-TF − ethics (pooled)	+0.068 [+0.010, +0.127]*	+0.110 [+0.005, +0.203]*
MMLU-TF − ETH-CS	+0.024 [−0.044, +0.094]	+0.045 [−0.065, +0.164]
MMLU-TF − ETH-JUSTICE	+0.109 [+0.040, +0.182]**	+0.171 [+0.062, +0.293]***

Table 5: Differences in AUROC and r between groups of tasks (two-sample bootstrap, 2,000 resamples; $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$). The binary factual control lies roughly halfway between the factual tasks and the pooled ethics tasks. It is indistinguishable from ETH-CS but clearly above ETH-JUSTICE.

Dataset	Best model	Vote − best [95% CI]	McNemar p (Holm)	vs. half-split best	Mean 5-subset vote − best	One per family
TRIVIAQA	gemma2-9b (0.725)	−2.4 [−5.1, 0.0]	0.35	−2.3	−4.6	−1.6
MMLU	qwen3-8b (0.728)	−1.8 [−5.0, +1.6]	0.71	−1.4	−1.5	−0.6
GSM8K	qwen3-8b (0.928)	**+2.4** [+0.6, +4.0]	0.085	**+3.2**	**+1.5**	**+1.8**
MATH-500	qwen3-8b (0.756)	+2.0 [−0.8, +4.6]	0.55	+3.1	−0.7	+0.4
ETH-CS	gemma2-9b (0.830)	−4.8 [−7.6, −2.2]	**0.008**	−4.7	−2.9	−4.2
ETH-JUSTICE	gemma2-9b (0.788)	−4.2 [−7.2, −1.2]	0.067	−4.0	−2.7	−5.4
MMLU-TF	gemma2-9b (0.724)	−0.6 [−4.2, +3.0]	0.83	+1.5	+0.1	−1.0

Table 6: Plurality vote minus best single model, in percentage points. "Best" is chosen on the test set (an oracle choice that favors the baseline), on a random half of items (mean of 200 splits), within each of the 56 five-model subsets, or within a pool of one model per family (qwen3-8b removed). Largest gain and the only significant McNemar test after Holm correction in **bold**. The vote helps only on math and is consistently worse than the best model on ethics.

0.587 on MMLU-TF. Asking other models is thus a more useful label-free check than reading a single model's confidence, and this holds in every domain.

4.3 About half of the ethics gap is due to answer format

Binary tasks differ from open-ended ones in two ways that weaken agreement as a signal. With two possible answers, models often agree by chance, and agreement can take only five values (4/8 to 8/8). The binary factual control MMLU-TF lets us measure this effect directly (table 5). MMLU-TF reaches AUROC 0.735, which is 0.056 below the open and multiple-choice factual tasks ($p = 0.03$). It is also 0.068 above the pooled ethics tasks ($p = 0.02$). The control therefore sits about halfway between the two groups. Its AUROC does not differ from ETH-CS ($\Delta = +0.024$, $p = 0.48$), but it is clearly above ETH-JUSTICE ($\Delta = +0.109$, $p = 0.002$). Most of the commonsense-morality gap is thus explained by format, while justice judgements carry an additional, domain-specific weakness. The high-agreement-but-wrong rate tells the same story. It is 13.4% on MMLU-TF, close to 13.8% on ETH-CS, and lower than 19.4% on ETH-JUSTICE.

4.4 Majority voting rarely beats the best model

The vote helps on math only. Table 6 compares the 8-model plurality vote with the best single model. On math, the vote gains $+2.4$ pp on GSM8K (95% CI $[+0.6, +4.0]$) and $+2.0$ pp on MATH-500. Neither gain survives Holm correction (both $p \geq 0.085$). On factual QA the vote is *below* the best model, by -2.4 pp on TRIVIAQA and -1.8 pp on MMLU. These differences are not significant, but every robustness variant points the same way. When the best model is selected on held-out halves, the difference is -2.3 and -1.4 pp. Averaged over all 56 five-model subsets, it is -4.6 and -1.5 pp. Our hypothesis that five or more models gain 3–7 pp on factual QA is therefore not supported.

On ethics, the vote is significantly worse than the best model. The vote is 4.8 pp below Gemma-2 on ETH-CS (Holm $p = 0.008$) and 4.2 pp below it on ETH-JUSTICE ($p = 0.067$; bootstrap CI

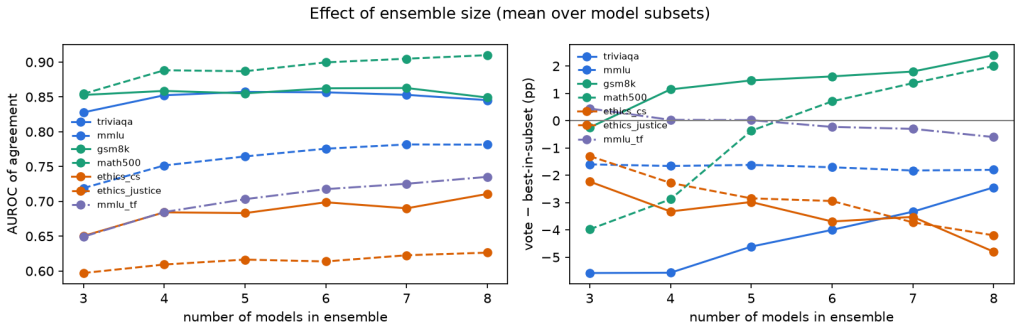


Figure 3: Effect of ensemble size k, averaged over all $\binom{8}{k}$ model subsets. **Left:** AUROC of agreement rises with k on every task, and the ranking of tasks stays stable. **Right:** vote minus best-in-subset accuracy. As models are added, the vote gains ground on math (green) and TRIVIAQA, but loses ground on both ethics tasks (orange), falling from -2.2/-1.3 pp at $k = 3$ to -4.8/-4.2 pp at $k = 8$.

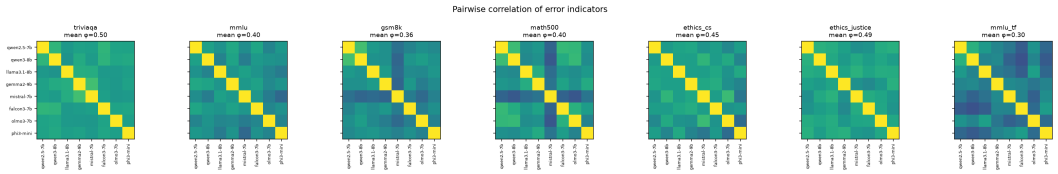


Figure 4: Pairwise ϕ correlation between the error indicators of the 8 models on each task (brighter means more correlated). Errors are more correlated on ethics (mean ϕ 0.45–0.49) than on the format-matched binary factual control (0.30).

$[-7.2, -1.2]$). These results hold when the best model is selected on held-out halves (-4.7 and -4.0 pp) and when we keep one model per family (-4.2 and -5.4 pp).

Adding models helps on math but hurts on ethics. Figure 3 varies the ensemble size from 3 to 8. More models always make agreement a better ranking signal: AUROC rises on every task, e.g., from 0.72 to 0.78 on MMLU and from 0.85 to 0.91 on MATH-500. The vote's accuracy relative to the best member, however, moves in opposite directions by domain. On math and TRIVIAQA, larger ensembles close the gap or turn it into a gain. MATH-500 goes from -4.0 to $+2.0$ pp, GSM8K from -0.3 to $+2.4$ pp, and TRIVIAQA from -5.6 to -2.4 pp. On ethics, larger ensembles make the vote *worse* relative to the best member. ETH-CS goes from -2.2 to -4.8 pp and ETH-JUSTICE from -1.3 to -4.2 pp. MMLU stays flat at about -1.7 pp. A growing deficit is what we expect when added members repeat the same errors rather than correcting them.

4.5 Why consensus fails: error structure

Ethics errors are more correlated than format alone explains. Figure 4 shows the pairwise correlation of model errors. On ethics, mean ϕ is 0.45 (ETH-CS) and 0.49 (ETH-JUSTICE), against 0.30 on the binary factual control. This is the domain-specific part of the ethics gap: when one model errs on an ethics item, the others tend to err on the same item. In binary tasks, two wrong models must give the same wrong answer, so $P(\text{same wrong} \mid \text{both wrong}) \approx 1$ for all three binary tasks and carries no evidence. On open-ended tasks this quantity is informative. It is only 0.08 on MATH-500, compared with 0.15 on TRIVIAQA and 0.59 on MMLU. Wrong chains of thought scatter across many different wrong numbers, so agreement on a wrong math answer is rare. This explains why math has the highest AUROC.

Factual consensus errors are rare and often trace to grading or the gold label. Only 3 TRIVIAQA items and 4 MMLU items are unanimous but wrong. One TRIVIAQA case is a grading artifact ("Land Rover" vs. the gold "Jaguar Land Rover"). The other two are genuine shared misconceptions. All 8 models answer "Hong Kong" for the car registration code HKJ (gold: Jordan), and "Winston Churchill" for the prime minister who married Gladys Mary Baldwin (gold: Harold Wil-

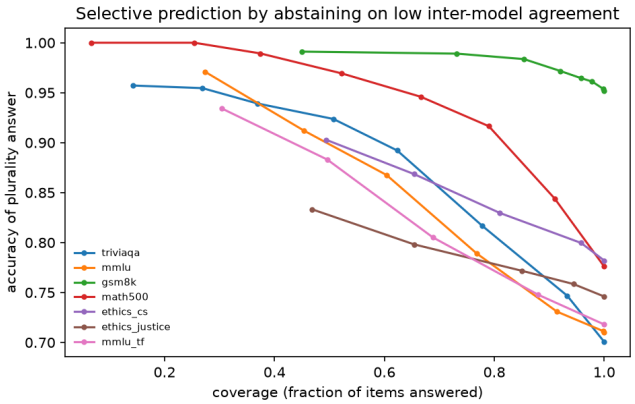


Figure 5: Coverage–accuracy curves when the system answers only items whose agreement is above a threshold. Curves for math and factual tasks rise steeply as coverage drops. The ETH-JUSTICE curve (brown) is nearly flat: abstaining on low-agreement items barely improves accuracy.

son). On MMLU, some cases come from questionable gold labels, e.g., all models define malware as "a virus or worm" while the gold answer is "a hacker tool".

Ethics consensus errors mix label noise with shared model leanings. There are 24 unanimous-but-wrong items on ETH-CS and 39 on ETH-JUSTICE. They fall into two groups. The first group consists of arguably debatable crowd labels. For example, all 8 models judge *"I took games away from my son as a punishment for skipping school"* and *"I buried her up to her neck on the beach"* as wrong, while the gold label is acceptable. The second group reflects systematic model leanings. Models call scenarios that involve harm to children, punishment or retribution "wrong" or "unreasonable" regardless of context. On justice, models accept weak excuses in the template "I usually … but I didn't today because …". For instance, all 8 models judge *"… pay for my son's lunch … but today I didn't because he took the bus to meet me"* as reasonable, while the gold label is unreasonable.

4.6 Agreement as a selective-prediction rule

Figure 5 uses agreement as an abstention rule. If we answer only when at least 6 of 8 models agree, accuracy on TRIVIAQA rises from 0.70 to 0.94 at 37% coverage, on MATH-500 from 0.78 to 0.99 at 37% coverage, and on MMLU from 0.71 to 0.87 at 60% coverage. On ethics the same rule keeps most items but gains little: ETH-CS rises from 0.78 to 0.83 at 81% coverage, and ETH-JUSTICE from 0.75 to 0.77 at 85% coverage.

4.7 Robustness

One model per family. Removing qwen3-8b leaves one model per developer family and does not change our conclusions (table 6, last column). AUROC is 0.84/0.78 on TRIVIAQA/MMLU, 0.87/0.89 on GSM8K/MATH-500 and 0.71/0.64 on ETH-CS/ETH-JUSTICE.

Rerun reproducibility. We reran Phi-3 on 50 items per task with a different batch composition. Parsed answers were identical for 98–100% of direct-answer items and 90% of GSM8K items, but only 48% of MATH-500 items. Long 4-bit chains of thought diverge numerically, mostly on items the model gets wrong. Per-task accuracy changed by at most 2 pp.

5 Discussion

Table 7 summarizes the verdict on each sub-hypothesis. We now discuss what the results mean.

Agreement is most informative when the answer space is large. When answers are open-ended, as in TRIVIAQA and MATH-500, independent models rarely produce the same wrong string or number by chance. Agreement on such tasks is strong evidence of a correct answer, and unanimous

Sub-hypothesis	Verdict	Evidence
H1a: $r \in [0.4, 0.6]$ on factual and math	Mostly supported	TRIVIAQA 0.55, MMLU 0.45, MATH-500 0.61 in range; GSM8K 0.37 (few errors limit variance; AUROC 0.85).
H1b: $r < 0.2$ on ethics	Partly supported	ETH-JUSTICE $r = 0.20$ $[0.11, 0.29]$ on the boundary; ETH-CS $r = 0.33$. Significantly below factual and math, but not near zero.
H2a: vote beats best by 3–7 pp on factual QA	Not supported	-2.4 and -1.8 pp (n.s.); gains of about 2 pp only on math.
H2b: vote below best on ethics	Supported	-4.8 pp (Holm $p = 0.008$) and -4.2 pp; the deficit grows with ensemble size.
H3: AUROC 0.70–0.80 on factual tasks	Supported (slightly exceeded)	MMLU 0.78, TRIVIAQA 0.85, pooled 0.79 $[0.76, 0.82]$.

Table 7: Verdict on each sub-hypothesis.

answers are right 96–100% of the time. Binary answer spaces remove much of this evidence. Models often agree by chance, and agreement can take only a few values. Comparisons of agreement across tasks should therefore account for the size of the answer space. Chance-corrected agreement measures are a natural next step.

Ethics is weaker for two distinct reasons. The first reason is format. The binary factual control loses 0.056 AUROC relative to open and multiple-choice factual tasks, and it matches ETH-CS. The second reason is specific to the domain. Errors are more correlated on ethics than on the control (ϕ 0.45–0.49 vs. 0.30), ETH-JUSTICE is significantly below the control, and consensus errors show shared leanings, such as treating any punishment as wrong. Our hypothesis attributed the whole ethics deficit to shared bias. The data suggest that this explains only about half of the gap for commonsense morality, but a larger part for justice. Part of the measured "error" on ethics is also disagreement with crowd labels that are themselves debatable.

Ranking by agreement and voting are different uses of consensus. Agreement ranks items well even where voting does not improve accuracy. On factual QA, knowledge is spread unevenly across models. The strongest model knows many facts that weaker models do not, and the plurality outvotes it. H2a implicitly assumed members of similar strength with independent errors. Our pool has neither property. In this regime, adding members dilutes the best model on factual tasks, and it reinforces shared mistakes on ethics. Math is the exception, because wrong chains of thought rarely land on the same answer.

Practical guidance. Cross-model agreement is a good label-free *triage* signal. Practitioners can accept unanimous or near-unanimous answers on factual and math tasks, and send low-agreement items to a stronger model or to a human. It does not replace using the best available model, because a plurality vote over a mixed pool is not a free accuracy gain. For moral judgements, consensus among models should not stand in for human labels. Between 10% and 17% of unanimous ETHICS verdicts contradict the crowd label, and adding models makes the vote worse relative to its best member. This matters for LLM-as-judge panels and synthetic preference data, where model consensus is often treated as ground truth.

5.1 Limitations

Model scale and precision. All models have 3.8–9B parameters and run at 4-bit precision, because only local open models on shared GPUs were available. Frontier models may be more accurate, more correlated with each other [Goel et al., 2025], or both. The absolute vote-versus-best numbers in particular may not transfer.

Pool composition. Each task has one clearly stronger model, and the vote-versus-best comparison depends strongly on this mix. The ensemble-size sweep and the one-per-family analysis address this only in part.

Ethics ground truth. ETHICS labels come from crowdworkers, and the hard test split is adversarially filtered. Some "errors" are label disagreements rather than model mistakes (section 4.5). Correctness on ethics therefore partly means agreement with the norms of US crowdworkers.

Post-hoc choices. We tightened the TRIVIAQA equivalence rule after inspecting errors, and we added the binary control after seeing the main results. Answer matching for TRIVIAQA and MATH-500 is heuristic, and spot checks reveal remaining misses (e.g., "Land Rover" vs. "Jaguar Land Rover"). Most such misses lower the measured accuracy of high-agreement items, which makes our factual AUROC estimates conservative.

Prompting and determinism. Each model gives one greedy answer, with direct answers on factual and ethics tasks. Chain of thought on ethics or MMLU might change the agreement structure, and we did not test prompt variants. Long 4-bit chains of thought also change with batch composition, so individual MATH-500 answers are not reproducible, even though accuracy is stable to within 2 pp.

Contamination and sample size. All benchmarks are public and may appear in training data. This affects absolute accuracy more than the relation between agreement and correctness. With 500 items per task, confidence intervals on r are about ± 0.07, and vote-versus-best differences of 2 pp cannot be resolved.

6 Conclusion

We studied how far agreement among independent LLMs predicts the correctness of their answers. We used 8 open-weight models from 7 families on factual, math and ethics tasks, plus a format-matched binary factual control. Agreement is a strong label-free correctness signal for factual (AUROC 0.78–0.85) and math answers (0.85–0.91), and a much weaker one for ethical judgements (0.63–0.71). Only part of the ethics gap reflects shared model bias. The rest comes from the binary answer format and from noise in the crowd labels. Plurality voting beats the best single model only on math. It is neutral to slightly harmful on factual QA, and significantly harmful on ethics, where the deficit grows with every model added, as correlated errors would predict. The key takeaway is that model consensus is a useful filter for facts and math, but not a substitute for human judgement on moral questions.

Several directions follow from this work. First, repeating the study with frontier API models would test whether stronger models are also more correlated. Second, chance-corrected agreement and binary versions of open-ended tasks would further separate format from domain. Third, moral-judgement datasets with several annotations per item, such as Scruples [Lourie et al., 2021] or MoralChoice [Scherrer et al., 2023], would let us test whether model agreement tracks *human* agreement rather than majority labels. Finally, accuracy-weighted voting, agreement combined with self-probability, and a comparison with within-model self-consistency at matched compute [Wang et al., 2023] could make consensus more reliable.

References

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Thomas G. Dietterich. Ensemble methods in machine learning. In *International Workshop on Multiple Classifier Systems*, pages 1–15, 2000.

Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations (ICLR)*, 2023.

Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Shashwat Goel, Joschka Struber, Ilze Amanda Auzina, Karuna K. Chandra, Ponnurangam Kumaraguru, Douwe Kiela, Ameya Prabhu, Matthias Bethge, and Jonas Geiping. Great models

think alike and this undermines AI oversight. In *International Conference on Machine Learning (ICML)*, 2025.

Dan Hendrycks, Collin Burns, Steven Basart, Andrew Critch, Jerry Li, Dawn Song, and Jacob Steinhardt. Aligning AI with shared human values. In *International Conference on Learning Representations (ICLR)*, 2021a.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations (ICLR)*, 2021b.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021c.

Dongfu Jiang, Xiang Ren, and Bill Yuchen Lin. LLM-Blender: Ensembling large language models with pairwise ranking and generative fusion. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

Mandar Joshi, Eunsol Choi, Daniel Weld, and Luke Zettlemoyer. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2017.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Elliot Kim, Avi Garg, Kenny Peng, and Nikhil Garg. Correlated errors in large language models. In *International Conference on Machine Learning (ICML)*, 2025.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations (ICLR)*, 2023.

Ludmila I. Kuncheva and Christopher J. Whitaker. Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy. *Machine Learning*, 51(2):181–207, 2003.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.

Junyou Li, Qin Zhang, Yangbin Yu, Qiang Fu, and Deheng Ye. More agents is all you need. *Transactions on Machine Learning Research*, 2024.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024.

Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration. In *Proceedings of Machine Learning and Systems (MLSys)*, 2024.

Nicholas Lourie, Ronan Le Bras, and Yejin Choi. Scruples: A corpus of community ethical judgments on 32,000 real-life anecdotes. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2021.

Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

Nino Scherrer, Claudia Shi, Amir Feder, and David Blei. Evaluating the moral beliefs encoded in LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Pat Verga, Sebastian Hofstatter, Sophia Althammer, Yixuan Su, Aleksandra Piktus, Arkady Arkhangorodsky, Minjie Xu, Naomi White, and Patrick Lewis. Replacing judges with juries: Evaluating LLM generations with a panel of diverse models. *arXiv preprint arXiv:2404.18796*, 2024.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

A Additional Details

Per-model accuracy. Figure 6 visualizes the accuracies in table 3.

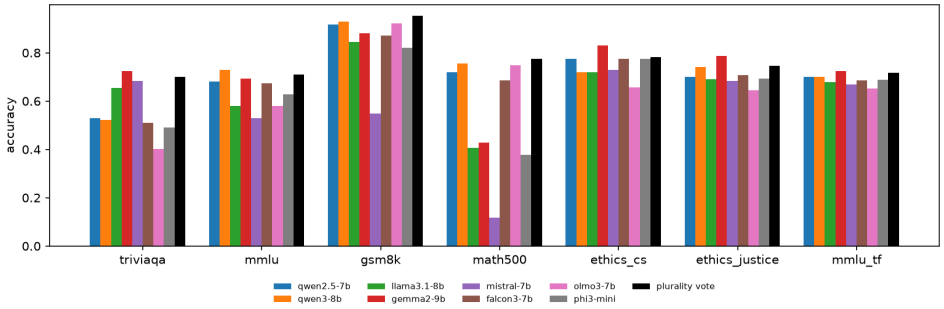


Figure 6: Accuracy of each model and of the 8-model plurality vote on each task.

Selective prediction at two thresholds. Table 8 lists coverage and accuracy when the system answers only items with agreement at or above a threshold.

		$a(x) \geq 0.75$		$a(x) = 1$	
Dataset	Acc. (all)	Coverage	Accuracy	Coverage	Accuracy
TRIVIAQA	0.701	0.369	0.939	0.143	0.957
MMLU	0.710	0.604	0.868	0.274	0.971
GSM8K	0.952	0.854	0.984	0.450	0.991
MATH-500	0.776	0.374	0.989	0.066	1.000
ETH-CS	0.782	0.810	0.830	0.494	0.903
ETH-JUSTICE	0.746	0.850	0.772	0.468	0.833
MMLU-TF	0.718	0.688	0.805	0.304	0.934

Table 8: Selective prediction using inter-model agreement as the abstention rule.

One model per family. Table 9 repeats the main analysis without qwen3-8b, so that each developer family contributes one model (7 models).

Implementation. We used vLLM 0.31.0 with a maximum context length of 2,560 tokens. A scheduler waited for free GPU memory and retried runs that failed at startup. Every rendered prompt, all raw generations and all parsed answers are stored with the code. The API cost was zero.

Dataset	**Best model**	Vote acc.	Vote − best (pp)	r	AUROC
TRIVIAQA	gemma2-9b (0.725)	0.709	-1.6	0.538	0.841
MMLU	gemma2-9b (0.692)	0.686	-0.6	0.462	0.784
GSM8K	olmo3-7b (0.922)	0.940	$+1.8$	0.438	0.869
MATH-500	olmo3-7b (0.748)	0.752	$+0.4$	0.587	0.891
ETH-CS	gemma2-9b (0.830)	0.788	-4.2	0.321	0.711
ETH-JUSTICE	gemma2-9b (0.788)	0.734	-5.4	0.223	0.640
MMLU-TF	gemma2-9b (0.724)	0.714	-1.0	0.366	0.727

Table 9: Results with one model per family. The conclusions match the full pool.